

Critical packings, rigidity, and the radius function

Wöden Kusner

Institute for Analysis and Number Theory
Graz University of Technology



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Abstract

There are a number of classical problems in geometric optimization that ask for the "best" configuration of points with respect to some function. We are interested in the relationships between various notions of criticality for such functions on configuration spaces, in particular the injectivity or packing radius. This is not a Morse function, but it has been observed to be Morse-like, in that the topological notion of regularity can be defined in an analogous way. Furthermore, there is a geometric interpretation from rigidity theory that characterizes configurations as critical by the existence of a strut measure.

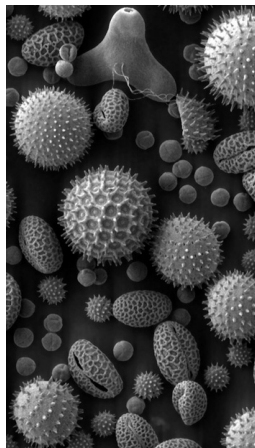
Based on work with
Robert Kusner, Jeffrey Lagarias & Senya Shlosman
arXiv 1611.10297

Tammes Problem: P. M. L. Tammes (1930)

Question

What is the maximal radius possible for N equal spheres, all touching a central sphere of radius 1?

The original formulation of the *Tammes problem*: How many spherical caps of angular diameter θ that can be placed without overlap?

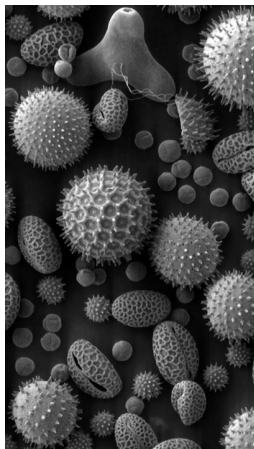


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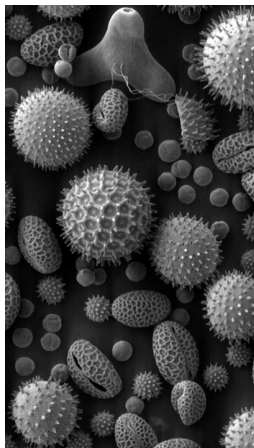


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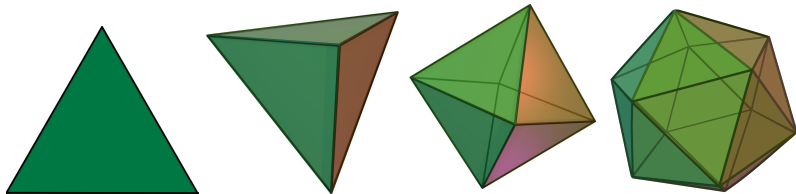


Remark

The maximizing configuration for 5 is not unique.

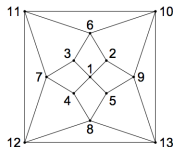
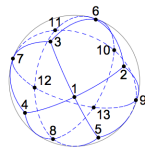
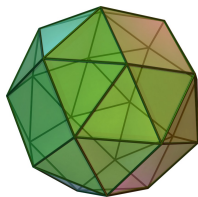
Tammes Problem: L. Fejes-Tóth (1943)

The Tammes problem was initially solved for $N = 3, 4, 6$ and 12 , with configurations of cap centers for $N = 3$ attained by vertices of an equatorial equilateral triangle and for $N = \{4, 6, 12\}$ by vertices of regular tetrahedron, octahedron and icosahedron.



Tammes Problem: Other N

The Tammes problem has been solved exactly for only $3 \leq N \leq 14$ and $N = 24$. It was solved for $N = \{5, 7, 8, 9\}$ by Schütte and van der Waerden in 1951, $N = \{10, 11\}$ by Danzer in his 1963 Habilitationsschrift.



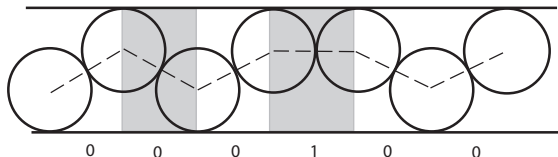
The case $N = 24$ was solved by Robinson in 1961 showing the configuration of centers were the vertices of a snub cube. The cases $N = \{13, 14\}$ were solved by Musin and Tarasov, enumerating all plausible graphs by computer.

Question

We have some solutions for the global maxima for the Tammes Problem, but there could be other interesting configurations. What about other critical points? Are there local maxima?

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Remark

One “similar” model that can be analyzed completely is the quasi-1D packing problem. Such packings have lots of maxima.

Definition

The classical *configuration space* $\text{Conf}(N) := \text{Conf}(\mathbb{S}^2, N)$ of N distinct labeled points on the unit 2-sphere $\mathbb{S}^2$.

Definition

The *reduced configuration space*

$$\text{BConf}(N) := \text{Conf}(N)/SO(3).$$

Assume $N \geq 3$ to avoid degenerate cases.

Definition

Configurations are $\mathbf{U} := (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_N)$, where the $\mathbf{u}_j \in \mathbb{S}^2$ are distinct points.

Definition

The *injectivity radius function* $\rho : \text{Conf}(N) \rightarrow \mathbb{R}^+$ assigns a configuration $\mathbf{U} := (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_N) \in (\mathbb{S}^2)^N$ the value

$$\rho(\mathbf{U}) := \frac{1}{2} \left(\min_{i \neq j} \theta(\mathbf{u}_i, \mathbf{u}_j) \right),$$

where $\theta(\mathbf{u}_i, \mathbf{u}_j)$ is the angular distance between $\mathbf{u}_i$ and $\mathbf{u}_j$.

Remark

Since ρ is invariant under the action of $SO(3)$, it descends to a well-defined function on $\text{BConf}(N; \theta)$, which we also denote ρ .

Definition

$$\text{Conf}(N; \theta) := \{\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_N) : \rho(\mathbf{U}) \geq \frac{\theta}{2}\}.$$

Morse theory concerns how topology changes for the super level sets of a smooth real-valued function on a manifold.

Definition (super level set)

For $f : M \rightarrow \mathbb{R}$, $M^a := \{x \in M : f(x) \geq a\}$ is a superlevel set.

Theorem

Given a smooth function $f : M \rightarrow \mathbb{R}$ and an interval $[a, b]$ with compact preimage, and $[a, b]$ contains no critical values. Then M^a is diffeomorphic to M^b .

It is at the *critical values* of the function that the topology of the super level changes.

To vary a configuration $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_N) \in \text{Conf}(N) \subset (\mathbb{S}^2)^N$ along a tangent vector $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_N)$ to $\text{Conf}(N)$ at $\mathbf{U}$, use an ersatz exponential map.

For sufficiently small $\mathbf{V}$, define a nearby configuration

$$\mathbf{U} \# \mathbf{V} = \left(\frac{\mathbf{u}_1 + \mathbf{v}_1}{|\mathbf{u}_1 + \mathbf{v}_1|}, \dots, \frac{\mathbf{u}_N + \mathbf{v}_N}{|\mathbf{u}_N + \mathbf{v}_N|} \right) \in \text{Conf}(N) \subset (\mathbb{S}^2)^N$$

by summing and projecting each factor back to $\mathbb{S}^2$.

In particular, the $\mathbf{V}$ -directional derivative of a smooth function f on $\text{Conf}(N)$ at $\mathbf{U}$ is $\frac{d}{dt}|_{t=0} f(\mathbf{U} \# t\mathbf{V})$.

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Definition

$\mathbf{U}$ is a critical point for smooth f provided all its $\mathbf{V}$ -derivatives vanish at $\mathbf{U}$. That is, the increment $f(\mathbf{U} \# \mathbf{V}) - f(\mathbf{U}) = o(\mathbf{V})$.

Remark

The injectivity radius function is not Morse.

The injectivity radius function ρ on $\text{Conf}(N)$ is not smooth: it is a min-function for a finite number of smooth functions. But we may still be inspired by Morse theory to pass between the topological, analytic and geometric notions of “critical”.

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Definition

A configuration $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_N) \in \text{Conf}(N)$ is *critical for maximizing* ρ provided for every sufficiently small $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_N)$, we have $\max[\rho(\mathbf{U} \# \mathbf{V}) - \rho(\mathbf{U}), 0] = o(\|\mathbf{V}\|)$. That is, a configuration $\mathbf{U}$ is *critical for maximizing* if there is no variation $\mathbf{V}$ that can increase ρ to first order.

If we are not critical for maximizing, there exists a variation $\mathbf{V}$ which increases ρ to first order. By the definition of ρ as a min-function, the distance between all pairs $(\mathbf{u}_i, \mathbf{u}_j)$ realizing the minimal angular distance $\theta(\mathbf{u}_i, \mathbf{u}_j) = \theta_o$ increases to first order along $\mathbf{V}$. So there is also a notion of regular value analogous to the smooth case.

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Theorem (Topological Regularity)

If such a variation exists for all configurations in this $\rho = \theta_o$ -level set, then this level is topologically regular: that is, the variations provide a deformation retraction from $\text{Conf}(N; \theta_o - \varepsilon)$ to $\text{Conf}(N; \theta_o + \varepsilon)$ for some $\varepsilon > 0$.

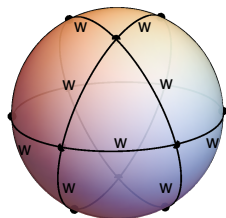
Balanced Graphs

Definition

For $\mathbf{U} \in \text{Conf}(N; \theta)$, the *contact graph* of $\mathbf{U}$ is the graph embedded in $\mathbb{S}^2$ with vertices given by points $\mathbf{u}_i$ in $\mathbf{U}$ and edges given by the geodesic segments $[\mathbf{u}_i, \mathbf{u}_j]$ when $d(\mathbf{u}_i, \mathbf{u}_j) = \theta$.

Definition

A *stress graph* for $\mathbf{U} \in \text{Conf}(N; \theta)$ is a contact graph with nonnegative weights w_e on each geodesic edge $e = [\mathbf{u}_i, \mathbf{u}_j]$.

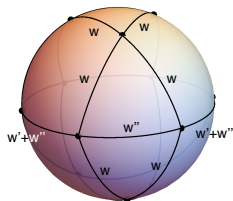


Balanced Graphs

A stress graph associates a system of *tangential forces* to edges $e = [\mathbf{u}_i, \mathbf{u}_j]$ of the contact graph. The forces have magnitude w_e , are tangent to $\mathbb{S}^2$ at each point $\mathbf{u}_i$ of $\mathbf{U}$, and point outward at the ends of each edge.

Definition

A stress graph is *balanced* if the sum of the forces in $T_{\mathbf{u}_i}\mathbb{S}^2$ is zero for all points of $\mathbf{U}$. A configuration $\mathbf{U}$ is *balanced* if it has a balanced stress graph for some choice of non-negative, not everywhere zero weights on its edges.



Theorem

For each topologically critical value θ for the injectivity radius ρ , there exists a balanced configuration $\mathbb{U}$. The vertices of the contact graph are a subset of the points in $\mathbb{U}$ and the geodesic edges of the contact graph all have length θ .

Theorem

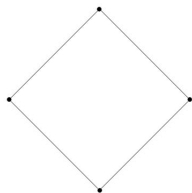
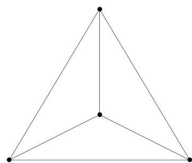
If a configuration $\mathbb{U}$ on $\mathbb{S}^2$ is balanced, then $\mathbb{U}$ is critical for maximizing the injectivity radius ρ .

Summary

There are certain radii that are critical:
The topology of the configuration space changes.

These radii also correspond to configurations of points that are force balanced: There exists a non-trivial strut measure on the contact graph that force balances all the vertices.

Such configurations obstruct the ρ -subgradient flow, which would give a deformation retraction.



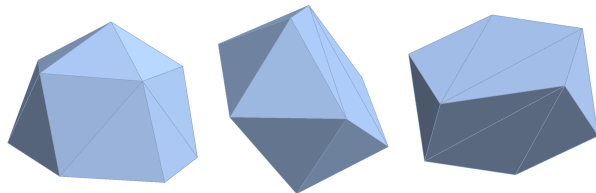
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We can hack together a sub-gradient descent algorithm for the injectivity radius function. With that, it is a quick step to make some conjectures about local maxima, global maxima, and the distribution of maxima just by exploring the basins of attraction randomly.



Thank you for your attention!

wkusner.github.io

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